# Indian Institute of Technology Madras

## Centre for Outreach and Digital Education



This is to certify that

**SHREY KUMAR SINGH**

has successfully completed the **Foundational Level** in

**Programming and Data Science**

M. Vignesh
**Coordinator**
IITM BS Program

V. Kamakoti
**Director**
IIT Madras

**Chair**
CODE, IIT Madras

# Indian Institute of Technology Madras

## PROGRESS CARD



**Name: SHREY KUMAR SINGH**
**Roll Number : 23F1002716**
**BS in Data Science and Applications**
**Current Level: FOUNDATION**

| Term | Course | Title | Credit | Category | Grade |
|---|---|---|---|---|---|
| SEP 2024 | HS1001 | English I | 4 | FL | S |
| SEP 2024 | CS1001 | Computational Thinking | 4 | FL | S |
| SEP 2024 | MA1001 | Mathematics for Data Science I | 4 | FL | D |
| SEP 2024 | MA1002 | Statistics for Data Science I | 4 | FL | C |

| S.no | COURSE | CGPA (max obtained) | CGPA (obtained) | GRADE |
|---|---|---|---|---|
| 1 | English I | 10 | 10 | S |
| 2 | Computational thinking | 10 | 10 | S |
| 3 | Mathematics for Data Science I | 10 | 6 | D |
| 4 | Statistics for Data Science I | 10 | 7 | C |

Cumulative Grade Point Average (CGPA) Summary

| Category | Foundation Level (FL) | Diploma - Programming (DP) | Diploma - Data Science (DD) | Degree - Programming (BP) | Degree-Data Science (BD) | Humanities & Management (HM) | Science and Engineering (SE) | Total |
|---|---|---|---|---|---|---|---|---|
| Min.Required Credits | 32 | 23 | 23 | 16 | 16 | 8 | 0 | 142 |
| Earned Credits | 32 | 15 | 4 | 0 | 0 | 0 | 0 | 51 |
| Transferred Credits Φ | - | - | - | - | - | - | - | 0 |
| CGPA | 10.0 | 10.0 | 10.0 | 0 | 0 | 0 | 0 | 10.0 |

Φ Transfererred Credits are not considered for CGPA calculation. Transferred Credits + Earned Credits should meet the total credit requirement

**Coordinator**
**IIT Madras BS Program**

Disclaimer: This document was generated electronically

# Indian Institute of Technology Madras

## PROGRESS CARD



**Name: SHREY KUMAR SINGH**
**Roll Number : 23F1002716**
**BS in Data Science and Applications**
**Current Level: FOUNDATION**

| Term | Course | Title | Credit | Category | Grade |
|---|---|---|---|---|---|
| FEB 2025 | HS1002 | English II | 4 | FL | S |
| FEB 2025 | CS1002 | Programming in Python | 4 | FL | A |
| FEB 2025 | MA1003 | Mathematics for Data Science II | 4 | FL | B |
| FEB 2025 | MA1004 | Statistics for Data Science II | 4 | FL | B |

| S.no | COURSE | CGPA (max obtained) | CGPA (obtained) | GRADE |
|---|---|---|---|---|
| 1 | English II | 10 | 10 | S |
| 2 | Programming in Python | 10 | 9 | A |
| 3 | Mathematics for Data science II | 10 | 7 | B |
| 4 | Statistics for Data Science II | 10 | 7 | B |

Cumulative Grade Point Average (CGPA) Summary

| Category | Foundation Level (FL) | Diploma - Programming (DP) | Diploma - Data Science (DD) | Degree - Programming (BP) | Degree-Data Science (BD) | Humanities & Management (HM) | Science and Engineering (SE) | Total |
|---|---|---|---|---|---|---|---|---|
| Min.Required Credits | 32 | 23 | 23 | 16 | 16 | 8 | 0 | 142 |
| Earned Credits | 32 | 15 | 4 | 0 | 0 | 0 | 0 | 51 |
| Transferred Credits Φ | - | - | - | - | - | - | - | 0 |
| CGPA | 10.0 | 10.0 | 10.0 | 0 | 0 | 0 | 0 | 10.0 |

Φ Transfererred Credits are not considered for CGPA calculation. Transferred Credits + Earned Credits should meet the total credit requirement

**Coordinator**
**IIT Madras BS Program**

Date:11 March 2024
Disclaimer: This document was generated electronically

# Indian Institute of Technology Madras

## PROGRESS CARD



**Name: SHREY KUMAR SINGH**
**Roll Number : 23F1002716**
**BS in Data Science and Applications**
**Current Level: FOUNDATION**

| Term | Course | Title | Credit | Category | Grade |
|---|---|---|---|---|---|
| JUL 2025 | CS2001 | Database Management Systems | 4 | DP | S |
| JUL 2025 | CS2002 | Programming, Data Structures and Algorithms using Python | 4 | DP | S |

| S.no | Course | CGPA (max obtained) | CGPA (obtained) | GRADE |
|---|---|---|---|---|
| 1 | Database Management Systems | 10 | 10 | S |
| 2 | Programing Data structure and Algorithm using Python | 10 | 10 | S |

Cumulative Grade Point Average (CGPA) Summary

| Category | Foundation Level (FL) | Diploma - Programming (DP) | Diploma - Data Science (DD) | Degree - Programming (BP) | Degree-Data Science (BD) | Humanities & Management (HM) | Science and Engineering (SE) | Total |
|---|---|---|---|---|---|---|---|---|
| Min.Required Credits | 32 | 23 | 23 | 16 | 16 | 8 | 0 | 142 |
| Earned Credits | 32 | 15 | 4 | 0 | 0 | 0 | 0 | 51 |
| Transferred Credits Φ | - | - | - | - | - | - | - | 0 |
| CGPA | 10.0 | 10.0 | 10.0 | 0 | 0 | 0 | 0 | 10.0 |

Φ Transfererred Credits are not considered for CGPA calculation. Transferred Credits + Earned Credits should meet the total credit requirement

**Coordinator**
**IIT Madras BS Program**

Date:11 March 2024
Disclaimer: This document was generated electronically

# Indian Institute of Technology Madras

## PROGRESS CARD



**Name: SHREY KUMAR SINGH**
**Roll Number : 23F1002716**
**BS in Data Science and Applications**
**Current Level: FOUNDATION**

| Term | Course | Title | Credit | Category | Grade |
|---|---|---|---|---|---|
| NOV 2025 | CS2003 | Modern Application Development I | 4 | DP | A |
| NOV 2025 | SE2001 | System Commands | 3 | DP | S |
| NOV 2025 | CS2004 | Machine Learning Foundations | 4 | DD | D |

| S.no | COURSE | CGPA (max obtained) | CGPA (obtained) | GRADE |
|---|---|---|---|---|
| 1. | Modern Application Development I | 10 | 9 | A |
| 2. | System Commands | 10 | 10 | S |
| 3. | Machine Learning Foundations | 10 | 6 | D |

Cumulative Grade Point Average (CGPA) Summary

| Category | Foundation Level (FL) | Diploma - Programming (DP) | Diploma - Data Science (DD) | Degree - Programming (BP) | Degree-Data Science (BD) | Humanities & Management (HM) | Science and Engineering (SE) | Total |
|---|---|---|---|---|---|---|---|---|
| Min.Required Credits | 32 | 23 | 23 | 16 | 16 | 8 | 0 | 142 |
| Earned Credits | 32 | 15 | 4 | 0 | 0 | 0 | 0 | 51 |
| Transferred Credits Φ | - | - | - | - | - | - | - | 0 |
| CGPA | 10.0 | 10.0 | 10.0 | 0 | 0 | 0 | 0 | 10.0 |

Φ Transfererred Credits are not considered for CGPA calculation. Transferred Credits + Earned Credits should meet the total credit requirement

**Coordinator**
**IIT Madras BS Program**

Disclaimer: This document was generated electronically

# Indian Institute of Technology Madras

## PROGRESS CARD



**Name: SHREY KUMAR SINGH**
**Roll Number : 23F1002716**
**BS in Data Science and Applications**
**Current Level: FOUNDATION**

| Term | Course | Title | Credit | Category | Grade |
|---|---|---|---|---|---|
| MAR 2026 | MA1001 | Mathematics for Data Science I | 4 | FL | A |
| MAR 2026 | MA1002 | Statistics for Data Science I | 4 | FL | A |
| MAR 2026 | CS2004 | Machine learning foundation | 4 | FL | B |

| S.no | COURSE | CGPA (max obtained) | CGPA (obtained) | GRADE |
|---|---|---|---|---|
| 1. | Mathematics for Data Science I | 10 | 9 | A |
| 2. | Statistics for Data Science I | 10 | 9 | A |
| 3. | Machine Learning Foundations | 10 | 8 | B |

Cumulative Grade Point Average (CGPA) Summary

| Category | Foundation Level (FL) | Diploma - Programming (DP) | Diploma - Data Science (DD) | Degree - Programming (BP) | Degree-Data Science (BD) | Humanities & Management (HM) | Science and Engineering (SE) | Total |
|---|---|---|---|---|---|---|---|---|
| Min.Required Credits | 32 | 23 | 23 | 16 | 16 | 8 | 0 | 142 |
| Earned Credits | 32 | 15 | 4 | 0 | 0 | 0 | 0 | 51 |
| Transferred Credits Φ | - | - | - | - | - | - | - | 0 |
| CGPA | 10.0 | 10.0 | 10.0 | 0 | 0 | 0 | 0 | 10.0 |

Φ Transfererred Credits are not considered for CGPA calculation. Transferred Credits + Earned Credits should meet the total credit requirement

**Coordinator**
**IIT Madras BS Program**

Disclaimer: This document was generated electronically

# Indian Institute of Technology Madras

## PROGRESS CARD



**Name: SHREY KUMAR SINGH**
**Roll Number : 23F1002716**
**BS in Data Science and Applications**
**Current Level: FOUNDATION**

| Term | Course | Title | Credit | Category | Grade |
|---|---|---|---|---|---|
| AUG 2026 | BS1001 | Modern Application Development | 4 | FL | A |
| AUG 2026 | BS1002 | Database Managements | 4 | FL | A |
| AUG 2026 | CS2008 | Programing Support using JAVA | 4 | FL | B |
| AUG 2026 | PMS002 | Business Analytics | 4 | FL | S |

| S.no | COURSE | CGPA (max obtained) | CGPA (obtained) | GRADE |
|---|---|---|---|---|
| 1. | Modern Application Development | 10 | 9 | A |
| 2. | Database Managements | 10 | 9 | A |
| 3. | Programing Support using JAVA | 10 | 8 | B |
| 4 | Business Analytics | 10 | 10 | S |

Cumulative Grade Point Average (CGPA) Summary

| Category | Foundation Level (FL) | Diploma - Programming (DP) | Diploma - Data Science (DD) | Degree - Programming (BP) | Degree-Data Science (BD) | Humanities & Management (HM) | Science and Engineering (SE) | Total |
|---|---|---|---|---|---|---|---|---|
| Min.Required Credits | 32 | 23 | 23 | 16 | 16 | 8 | 0 | 142 |
| Earned Credits | 32 | 15 | 4 | 0 | 0 | 0 | 0 | 51 |
| Transferred Credits Φ | - | - | - | - | - | - | - | 0 |
| CGPA | 10.0 | 10.0 | 10.0 | 0 | 0 | 0 | 0 | 10.0 |

Φ Transfererred Credits are not considered for CGPA calculation. Transferred Credits + Earned Credits should meet the total credit requirement

**Coordinator**
**IIT Madras BS Program**

Disclaimer: This document was generated electronically

## BS in Data Science and Applications

| Grade | | Remarks |
|---|---|---|
| Code | Points | |
| S | 10 | --- |
| A | 9 | --- |
| B | 8 | --- |
| C | 7 | --- |
| D | 6 | --- |
| E | 4 | --- |
| U | 0 | --- |
| P | 0 | Pass |
| F | 0 | Fail |
| *WA / WQ* | 0 | Not eligible for Exam |
| I | 0 | Course Incomplete |

**Grades 'S' to 'E' and 'P' indicate successful completion of course.**

The Grade of course(s) under the Pass/Fail category are not included towards CGPA Calculation.

$$\text{CGPA} = \frac{\sum_i (C_i \times GP)}{\sum_i C_i}$$

where $C_i$ = credit of the course

GP = Grade point for that course, and

$\sum_i$ = is the sum over all registered courses succesfully cleared during all the semester including those in which the student obtained 'U' and 'W' grades but not cleared.